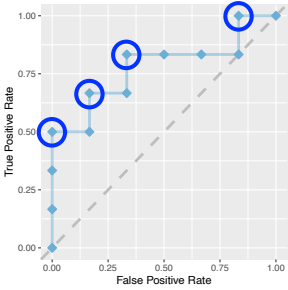
# Introduction to Machine Learning

# Evaluation
# Measures for Binary Classification: ROC Visualization

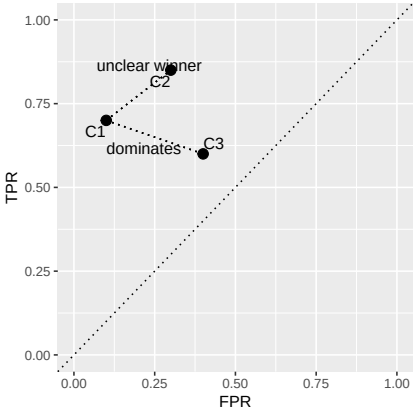



**Learning goals**

- Understand ROC curve
- Be able to compute a ROC curve manually
- Understand that ROC curve is invariant to class priors at test-time
- Discuss threshold selection
- Understand AUC

# LABELS: ROC SPACE

- For comparing classifiers, we characterize them by their TPR and FPR values and plot them in a coordinate system.
- We could also use two different ROC metrics which define a trade-off, for instance, TPR and PPV.




| | | **True Class** $y$ | |
|---|---|---|---|
| | | $+$ | $-$ |
| **Pred.** | $+$ | TP | FP |
| $\hat{y}$ | $-$ | FN | TN |

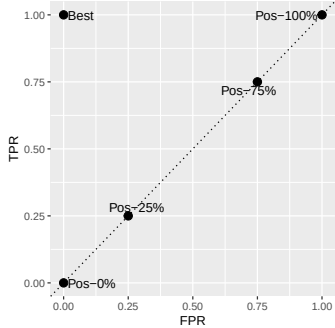
$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

# LABELS: ROC SPACE

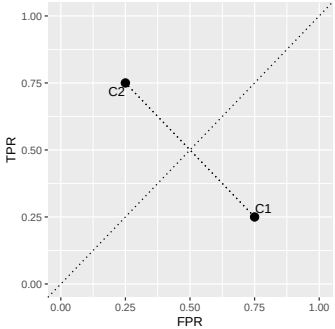
- The best classifier lies on the top-left corner, where FPR equals 0 and TPR is maximal.
- The diagonal is worst as it corresponds to a classifier producing random labels (with different proportions).

- If each positive $x$ will be randomly classified with 25% as "pos", TPR $= 0.25$.
- If we assign each negative $x$ randomly to "pos", FPR $= 0.25$.

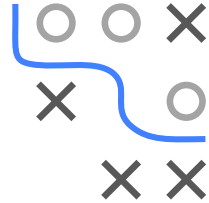


# LABELS: ROC SPACE

- In practice, we should never obtain a classifier below the diagonal.
- Inverting the predicted labels ($0 \mapsto 1$ and $1 \mapsto 0$) will result in a reflection at the diagonal.
  $\Rightarrow \text{TPR}_{\text{new}} = 1 - \text{TPR}$ and $\text{FPR}_{\text{new}} = 1 - \text{FPR}$.



# LABEL DISTRIBUTION IN TPR AND FPR



TPR and FPR (ROC curves) are insensitive to the class distribution in the sense that they are not affected by changes in the ratio $n_+/n_-$ (at prediction).

Example 1:
Proportion $n_+/n_- = 1$

| | Actual Positive | Actual Negative |
|---|---|---|
| Pred. Positive | 40 | 25 |
| Pred. Negative | 10 | 25 |

$\text{MCE} = 35/100 = 0.35$
$\text{TPR} = 0.8$
$\text{FPR} = 0.5$

Example 2:
Proportion $n_+/n_- = 2$

| | Actual Positive | Actual Negative |
|---|---|---|
| Pred. Positive | 80 | 25 |
| Pred. Negative | 20 | 25 |

$\text{MCE} = 45/150 = 0.3$
$\text{TPR} = 0.8$
$\text{FPR} = 0.5$

Note: If class proportions differ during training, the above is not true. Estimated posterior probabilities can change!

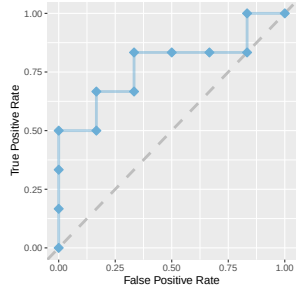
# FROM PROBABILITIES TO LABELS: ROC CURVE

Remember: Both probabilistic and scoring classifiers can output classes by thresholding:

$$h(\mathbf{x}) = [\pi(\mathbf{x}) \geq c] \quad \text{or} \quad h(\mathbf{x}) = [f(\mathbf{x}) \geq c_f].$$

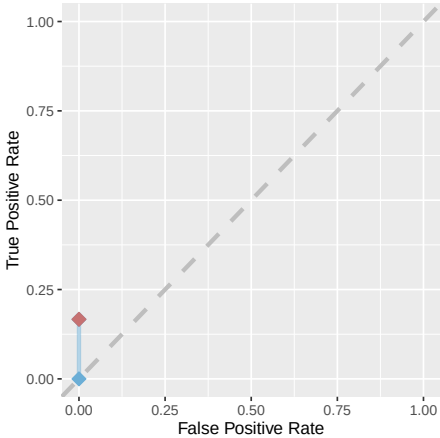
**To draw a ROC curve**:

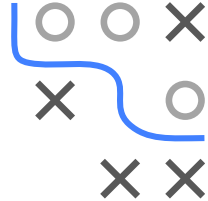
1. Rank test observations on decreasing score.
2. Start with $c = 1$, so we start in $(0, 0)$; we predict everything as negative.
3. Iterate through all possible thresholds $c$ and proceed for each observation $x$ as follows:
   - If $x$ is positive, move TPR $1/n_+$ up, as we have one TP more.
   - If $x$ is negative, move FPR $1/n_-$ right, as we have one FP more.



# DRAWING ROC CURVES

| # | Truth | Score |
|---|---|---|
| 1 | Pos | 0.95 |
| 2 | Pos | 0.86 |
| 3 | Pos | 0.69 |
| 4 | Neg | 0.65 |
| 5 | Pos | 0.59 |
| 6 | Neg | 0.52 |
| 7 | Pos | 0.51 |
| 8 | Neg | 0.39 |
| 9 | Neg | 0.28 |
| 10 | Neg | 0.18 |
| 11 | Pos | 0.15 |
| 12 | Neg | 0.06 |

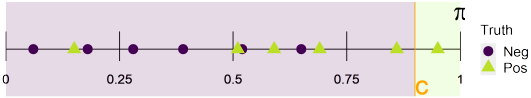


$c = 0.9$

$\rightarrow$ TPR = 0.167

$\rightarrow$ FPR = 0



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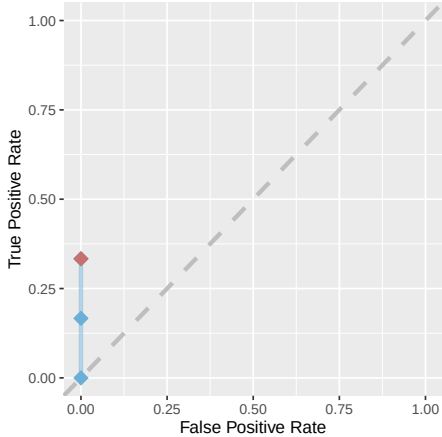
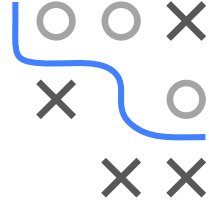
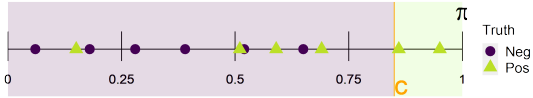


$c = 0.85$

$\rightarrow$ TPR = 0.333

$\rightarrow$ FPR = 0



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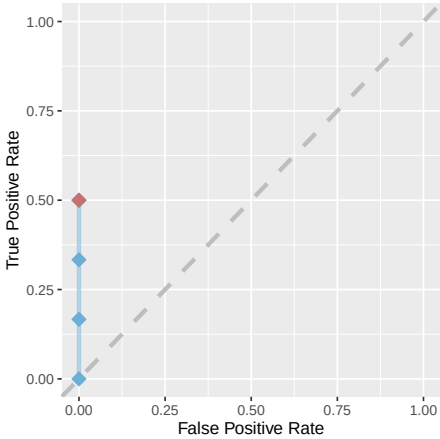
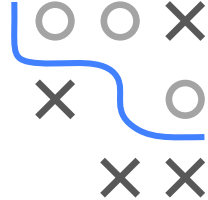
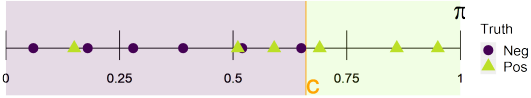


$c = 0.66$

$\rightarrow$ TPR = 0.5

$\rightarrow$ FPR = 0



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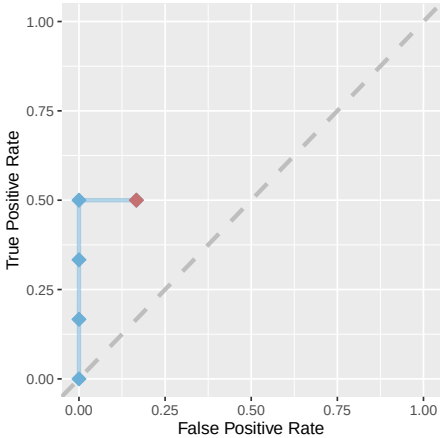
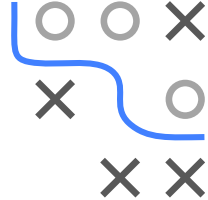
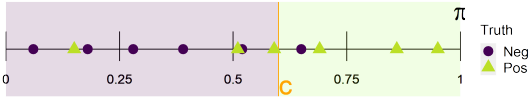


$c = 0.6$

$\rightarrow$ TPR = 0.5

$\rightarrow$ FPR = 0.167



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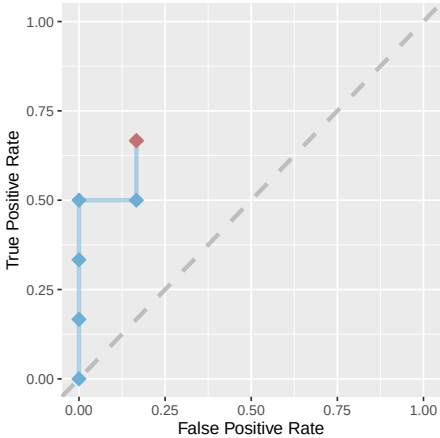
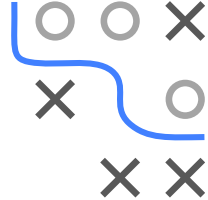
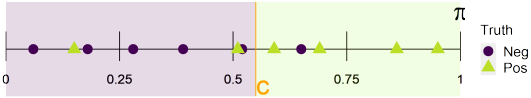


$c = 0.55$

$\rightarrow$ TPR = 0.667

$\rightarrow$ FPR = 0.167



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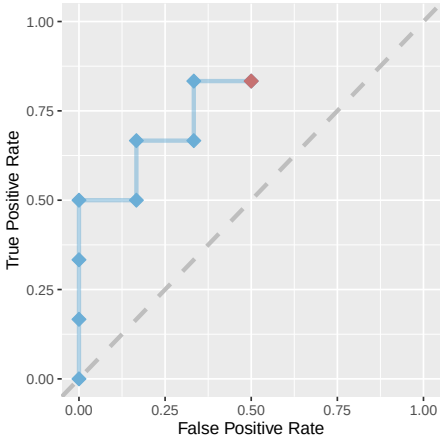
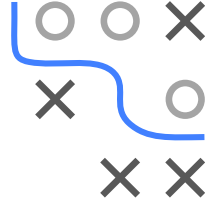
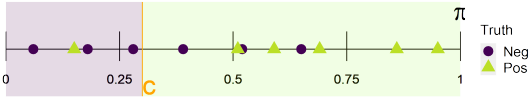


$c = 0.3$

$\rightarrow$ TPR = 0.833

$\rightarrow$ FPR = 0.5



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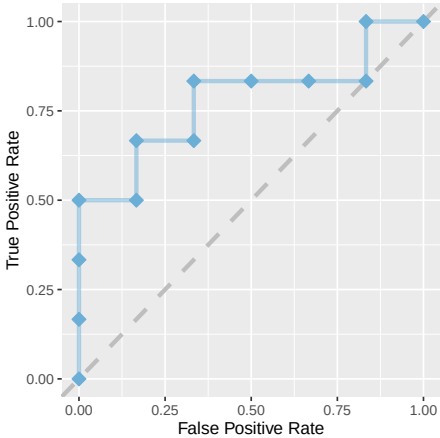
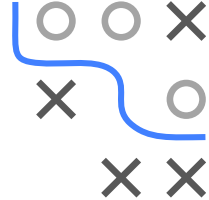
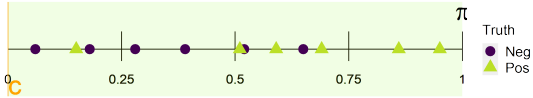


$c = 0$

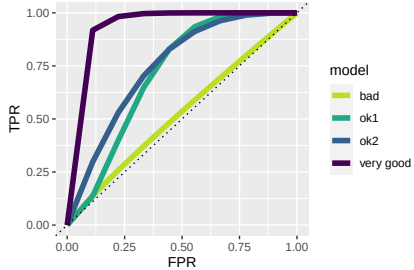
$\rightarrow$ TPR = 1

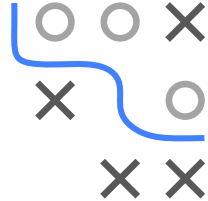
$\rightarrow$ FPR = 1



# ROC CURVE PROPERTIES

- The closer the curve to the top-left corner, the better.
- If ROC curves cross, a different model might be better in different parts of the ROC space.



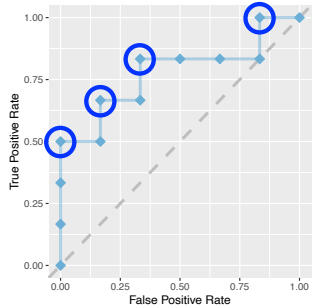



- Small thresholds will very liberally predict the positive class, and result in a potentially higher FPR, but also higher TPR.
- High thresholds will very conservatively predict the positive class, and result in a lower FPR and TPR.
- As we have not defined the trade-off between false positive and false negative costs, we cannot easily select the "best" threshold.
  $\rightarrow$ Visual inspection of all possible results seems useful.

# CHOOSING THRESHOLD / OPERATING POINT

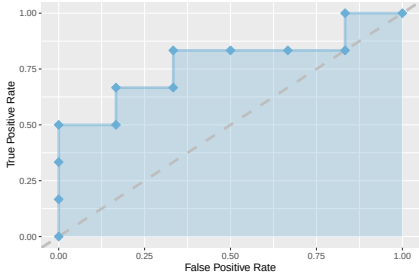
Often done visually and post-hoc, as class imbalances or costs are unknown a-priori.

- Identify non-dominated points
- Assess TPR / FPR
- Decide which combo is best for task
- Pick associated threshold



# AUC: AREA UNDER ROC CURVE

- AUC $\in [0, 1]$ is a single metric to evaluate scoring classifiers – independent of the chosen threshold.
  - AUC = 1: perfect classifier
  - AUC = 0.5: random, non-discriminant classifier
  - AUC = 0: perfect, with inverted labels



# AUC AS A RANK-BASED METRIC

- We can also interpret the AUC as the probability of our classifier ranking a random positive observation higher than a random negative one.
- A perfect classifier will rank all positive above all negative observations, achieving AUC = 1.

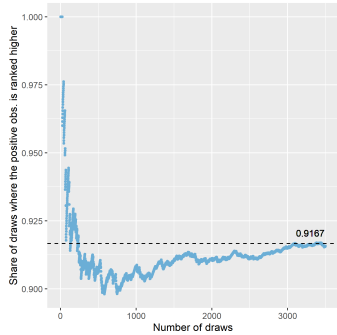
| Truth | Score |
|---|---|
| 1 | 0.9 |
| 1 | 0.76 |
| 1 | 0.7 |
| 0 | 0.5 |
| 1 | 0.45 |
| 0 | 0.3 |
| 0 | 0.1 |

AUC = 0.9167

Choose a random positive

| | |
|---|---|
| 1 | 0.76 |

Choose a random negative

| | |
|---|---|
| 0 | 0.3 |

- Classifier ranks the positive higher than the negative
- This happens with a mean probability of 0.9167